

Parcial I (Solución B)

(1)

$$1) a) \ln(f(x)) = \ln\left(\frac{1}{\beta} e^{-x/\beta}\right) \\ = -\ln(\beta) - \frac{x}{\beta}.$$

$$\Rightarrow \frac{\partial \ln(f(x))}{\partial \beta} = x\beta^{-2} - \frac{1}{\beta} = \frac{x - \beta}{\beta^2}$$

$$\Rightarrow E\left[\left(\frac{\partial \ln(f(x))}{\partial \beta}\right)^2\right] = \frac{1}{\beta^4} E[(x - \beta)^2] = \frac{1}{\beta^4} \text{Var}[X] = \frac{\beta^2}{\beta^4} = \frac{1}{\beta^2}.$$

$$\Rightarrow \text{Var}(\hat{\beta}) = \frac{\text{Var}[X]}{n} = \frac{\beta^2}{n} = \frac{1}{n E\left[\left(\frac{\partial \ln(f(x))}{\partial \beta}\right)^2\right]} = \frac{\beta^2}{n}$$

$\Rightarrow \hat{\beta}$ es EIVM de β .

$$2) a) E[X] = \frac{1}{2} \sqrt{\pi \theta} = \bar{x}$$

$\Rightarrow$ despejando θ para obtener el E.M de θ .

$$\Rightarrow \sqrt{\pi \theta} = 2\bar{x}$$

$$\Rightarrow \hat{\theta} = \left(\frac{2\bar{x}}{\sqrt{\pi}}\right)^2 \text{ es el EM de } \theta.$$

$$b) \mathcal{L}(\theta) = \frac{2^n \prod_{i=1}^n x_i}{\theta^n} e^{-\frac{\sum_{i=1}^n x_i^2}{\theta}}$$

$$\Rightarrow \ln(\mathcal{L}(\theta)) = \ell(\theta) = n \ln 2 - n \ln(\theta) + \ln\left(\prod_{i=1}^n x_i\right) - \frac{\sum_{i=1}^n x_i^2}{\theta}.$$

$$\Rightarrow \frac{\partial \ell(\theta)}{\partial \theta} = -\frac{1}{\theta} + \frac{\sum_{i=1}^n x_i^2}{\theta^2} = \frac{-\theta + \sum_{i=1}^n x_i^2}{\theta^2} = 0$$

$$\Rightarrow \hat{\theta} = \frac{\sum_{i=1}^n x_i^2}{n} \quad \text{es punto crítico}$$

$$\Rightarrow \frac{\partial^2 \ell(\theta)}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2 \sum_{i=1}^n x_i^2}{\theta^3}$$

$$\Rightarrow \left. \frac{\partial^2 \ell(\theta)}{\partial \theta^2} \right|_{\hat{\theta}} = \frac{n}{\left(\frac{\sum x_i^2}{n}\right)} - 2 \frac{\sum_{i=1}^n x_i^2}{\left(\frac{\sum x_i^2}{n}\right)^2} = \frac{-n^3}{\left(\sum x_i^2\right)^2} < 0.$$

$$\Rightarrow \hat{\theta} = \frac{\sum_{i=1}^n x_i^2}{n} \quad \text{es el EMV de } \theta.$$

c) $f(x)$ pertenece a la f. exponencial pues.

$$a(x) = x^2$$

$$b(x) = 2x$$

$$c(\theta) = \frac{1}{\theta}$$

$$d(\theta) = \frac{1}{\theta}.$$

$\Rightarrow \sum x_i^2$ es suficiente para θ .

$$\text{Tome } u = h(x) = x^2 \Rightarrow h^{-1}(u) = u^{\frac{1}{2}}, \Rightarrow \frac{\partial h^{-1}(u)}{\partial u} = \frac{1}{2} u^{-\frac{1}{2}}$$

$$\begin{aligned} \Rightarrow f_u(u) &= f(h^{-1}(u)) \left| \frac{\partial h^{-1}(u)}{\partial u} \right| \\ &= \frac{2u^{\frac{1}{2}}}{\theta} e^{-\frac{u}{\theta}} \left| \frac{1}{2u^{\frac{1}{2}}} \right| = \frac{1}{\theta} e^{-\frac{u}{\theta}}, \quad u > 0. \end{aligned}$$

$$\Rightarrow U \sim \text{Exp}(\theta)$$

13

$$\Rightarrow \sum_{i=1}^n X_i^2 \sim \text{Gamma}(n, \theta)$$

$$\Rightarrow E\left[\sum_{i=1}^n X_i^2\right] = n\theta \Rightarrow \hat{\theta} = \frac{\sum X_i^2}{n} \text{ es insesgado y}$$

función de un estadístico suficiente y suponiendo que es completo, por el T.L-S., $\hat{\theta}$ es EIVM de θ .

3)a) Sabemos que $X_0 \sim \text{Pareto}(\alpha n, \beta)$

$$\text{i.e. } f_{X_0}(x) = \begin{cases} \frac{n\alpha \beta^{\alpha n}}{x^{\alpha n+1}}, & x \geq \beta \\ 0, & x < \beta. \end{cases}$$

$$\text{Tome } U = \frac{X_0}{\beta} \Rightarrow h^{-1}(u) = \beta u \quad \text{y} \quad \frac{\partial h^{-1}(u)}{\partial u} = \beta.$$

$$\Rightarrow f_U(u) = f(h^{-1}(u)) \left| \frac{\partial h^{-1}(u)}{\partial u} \right| \\ = \frac{n\alpha}{u^{\alpha n+1}}, \quad u \geq \beta.$$

$$\Rightarrow U \sim \text{Pareto}(\alpha n, 1)$$

$\therefore U = \frac{X_0}{\beta}$ es pivote pues no depende de β y su f.dist. es

$$F_U(u) = \begin{cases} 1 - \left(\frac{1}{u}\right)^{\alpha n}, & u \geq 1 \\ 0, & u < 1 \end{cases}$$

$$b) P(a \leq U \leq b) = 1 - \lambda$$

$$P(U \leq a) = P(U \geq b) = \frac{\lambda}{2}$$

$$\Rightarrow 1 - \left(\frac{1}{a}\right)^{\alpha n} = \frac{\lambda}{2} \quad \gamma \quad 1 - \left(\frac{1}{b}\right)^{\alpha n} = 1 - \lambda$$

$$\Rightarrow a = \left(\frac{1}{1 - \frac{\lambda}{2}}\right)^{\frac{1}{\alpha n}} \quad \gamma \quad b = \left(\frac{2}{\lambda}\right)^{\frac{1}{\alpha n}}$$

$$= \left(\frac{2}{2 - \lambda}\right)^{\frac{1}{\alpha n}}$$

$$\Rightarrow P\left(\left(\frac{2}{2 - \lambda}\right)^{\frac{1}{\alpha n}} \leq \frac{X_{(1)}}{\beta} \leq \left(\frac{2}{\lambda}\right)^{\frac{1}{\alpha n}}\right) = 1 - \lambda$$

$$\Rightarrow P\left(\frac{X_{(1)}}{\left(\frac{2}{\lambda}\right)^{\frac{1}{\alpha n}}} \leq \beta \leq \frac{X_{(1)}}{\left(\frac{2}{2 - \lambda}\right)^{\frac{1}{\alpha n}}}\right) = 1 - \lambda$$

$\therefore \left(\frac{X_{(1)}}{\left(\frac{2}{\lambda}\right)^{\frac{1}{\alpha n}}}, \frac{X_{(1)}}{\left(\frac{2}{2 - \lambda}\right)^{\frac{1}{\alpha n}}}\right)$ es el IC bilateral de $(1 - \lambda)\%$ para β .